



Hemodynamic Disorders

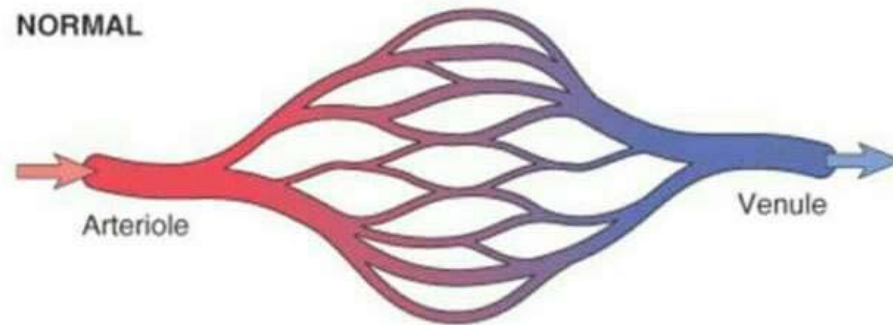
Lec. 7

Dr.Zainab Hashim

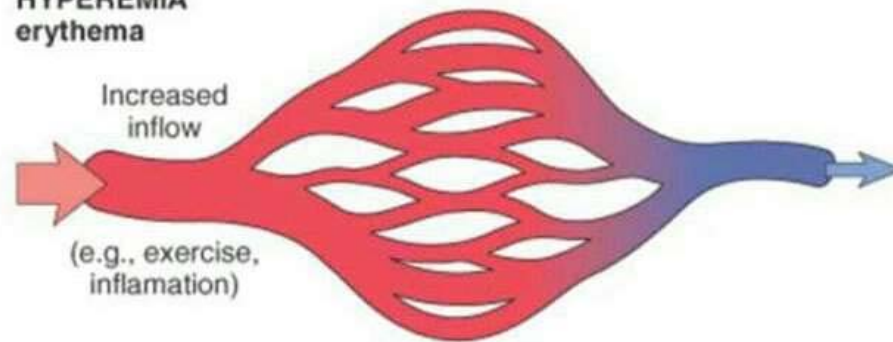
Hyperemia And Congestion

- Both refer to an increase in blood volume within a tissue, but have different underlying mechanisms.
- **Hyperemia** is an active process resulting from arteriolar dilation and increased blood inflow, as occurs at sites of inflammation or in exercising skeletal muscle. Hyperemic tissues are redder than normal because of engorgement with oxygenated blood.

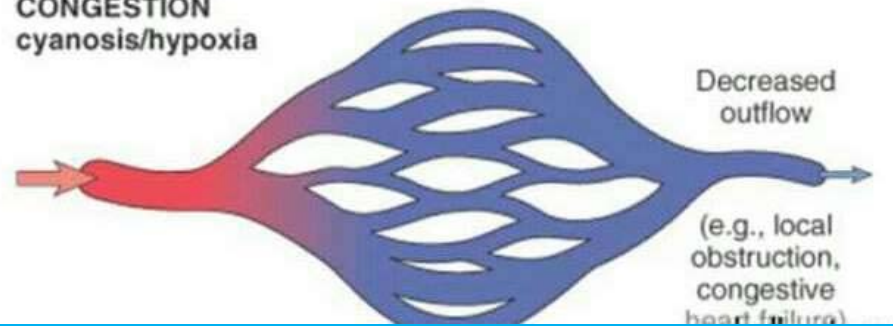
NORMAL



HYPEREMIA erythema



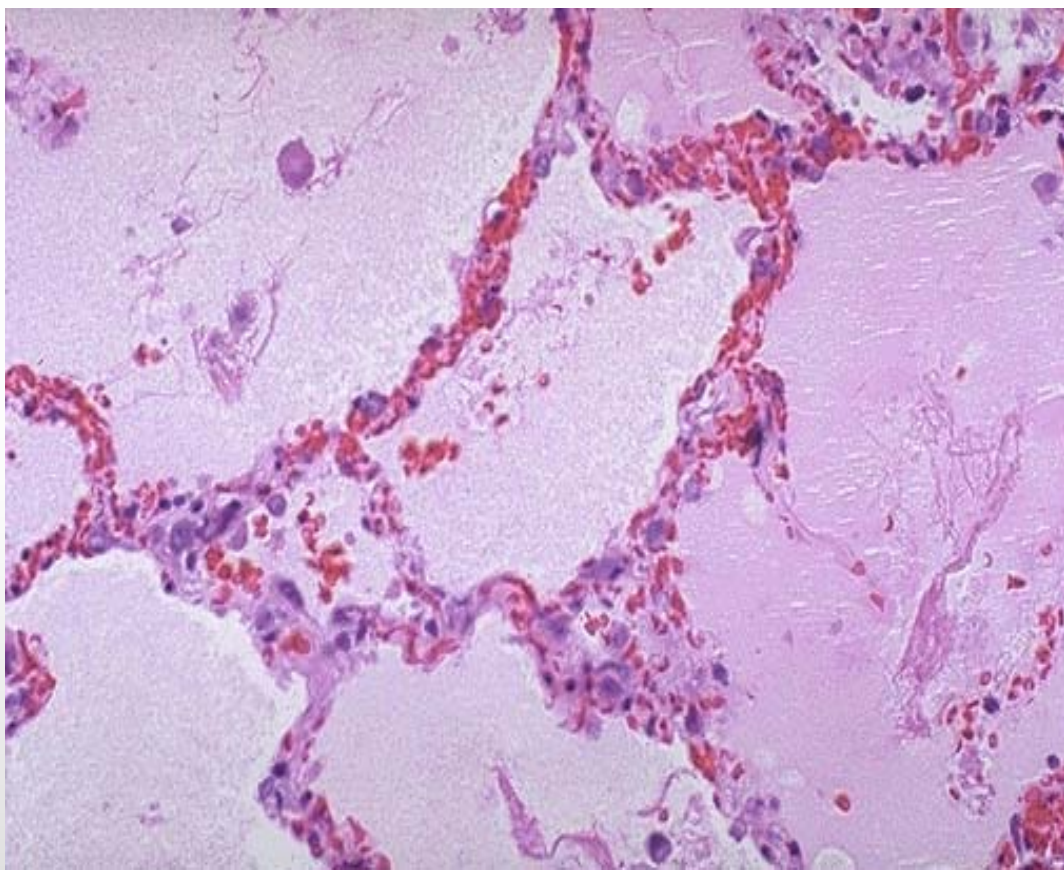
CONGESTION cyanosis/hypoxia



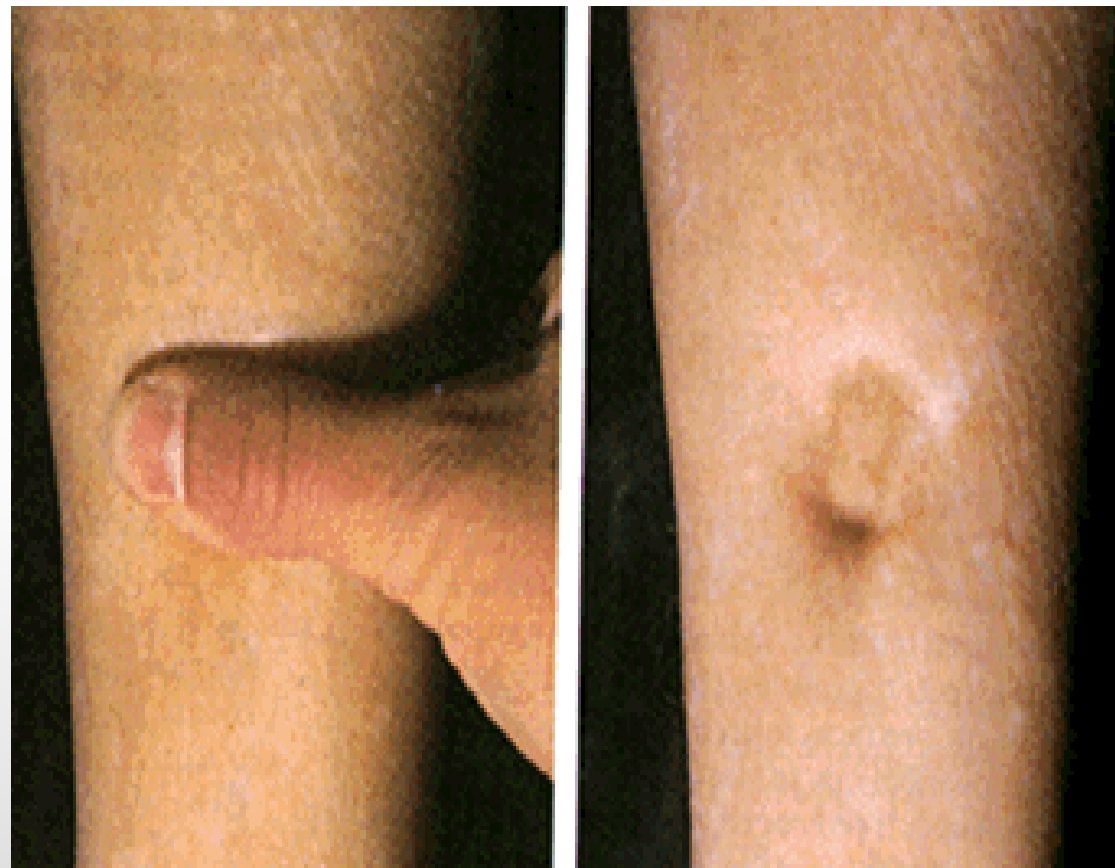
- **Congestion** is a passive process resulting from impaired outflow of venous blood from a tissue. It can occur systemically, as in cardiac failure, or locally as a consequence of an isolated venous obstruction. Congested tissues have an abnormal blue-red color (cyanosis) that stems from the accumulation of deoxygenated hemoglobin in the affected area.

Edema

- Edema is an accumulation of interstitial fluid within tissues. It results from the movement of fluid from the vasculature into the interstitial spaces.
- Extravascular fluid can also collect in body cavities and such accumulations are often referred to collectively as **effusions**.
- Examples include effusions in the pleural cavity (hydrothorax), the pericardial cavity (hydropericardium), or the peritoneal cavity (hydroperitoneum, or ascites).



Pulmonary edema : alveolar space filled with pale pink edema fluid

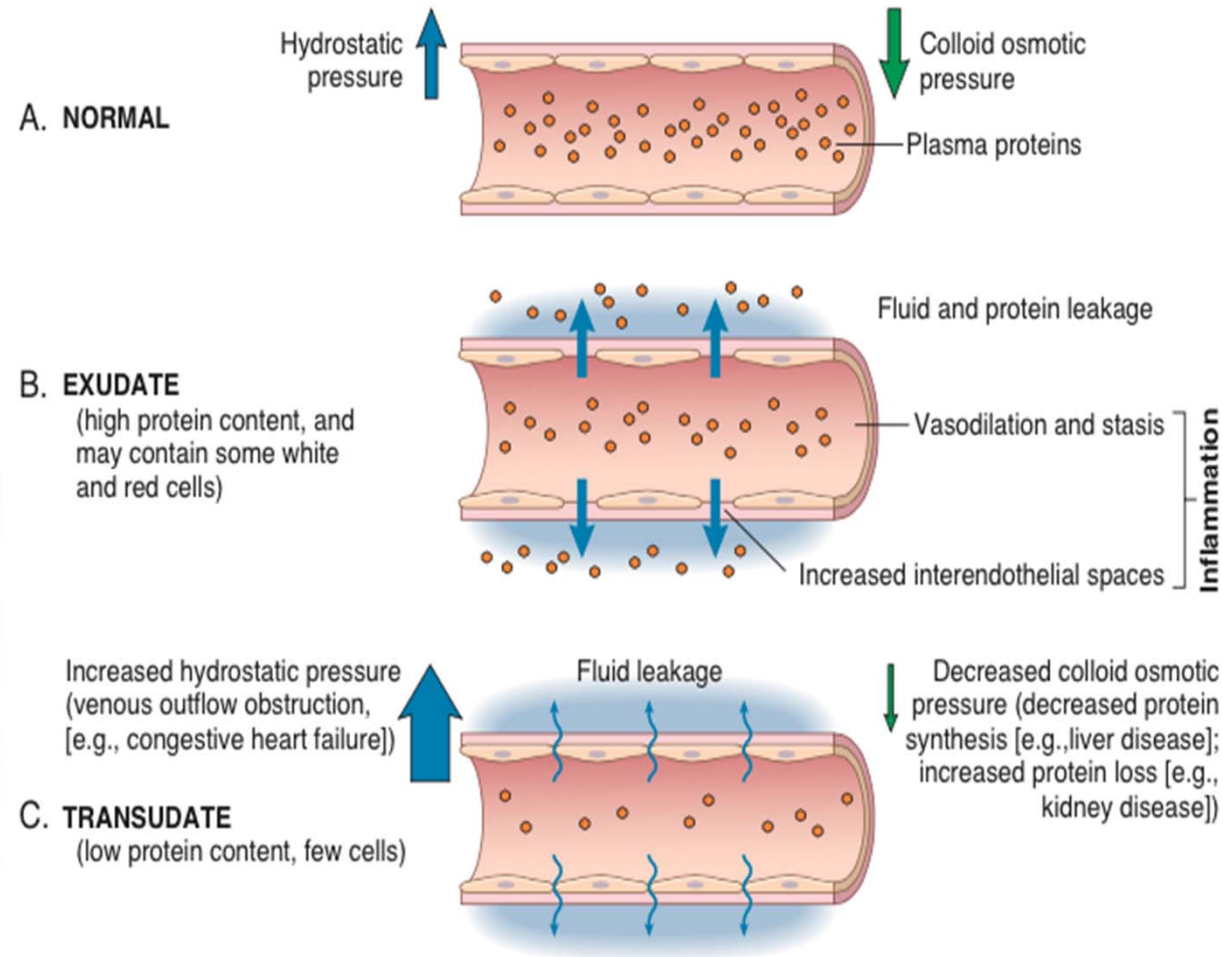


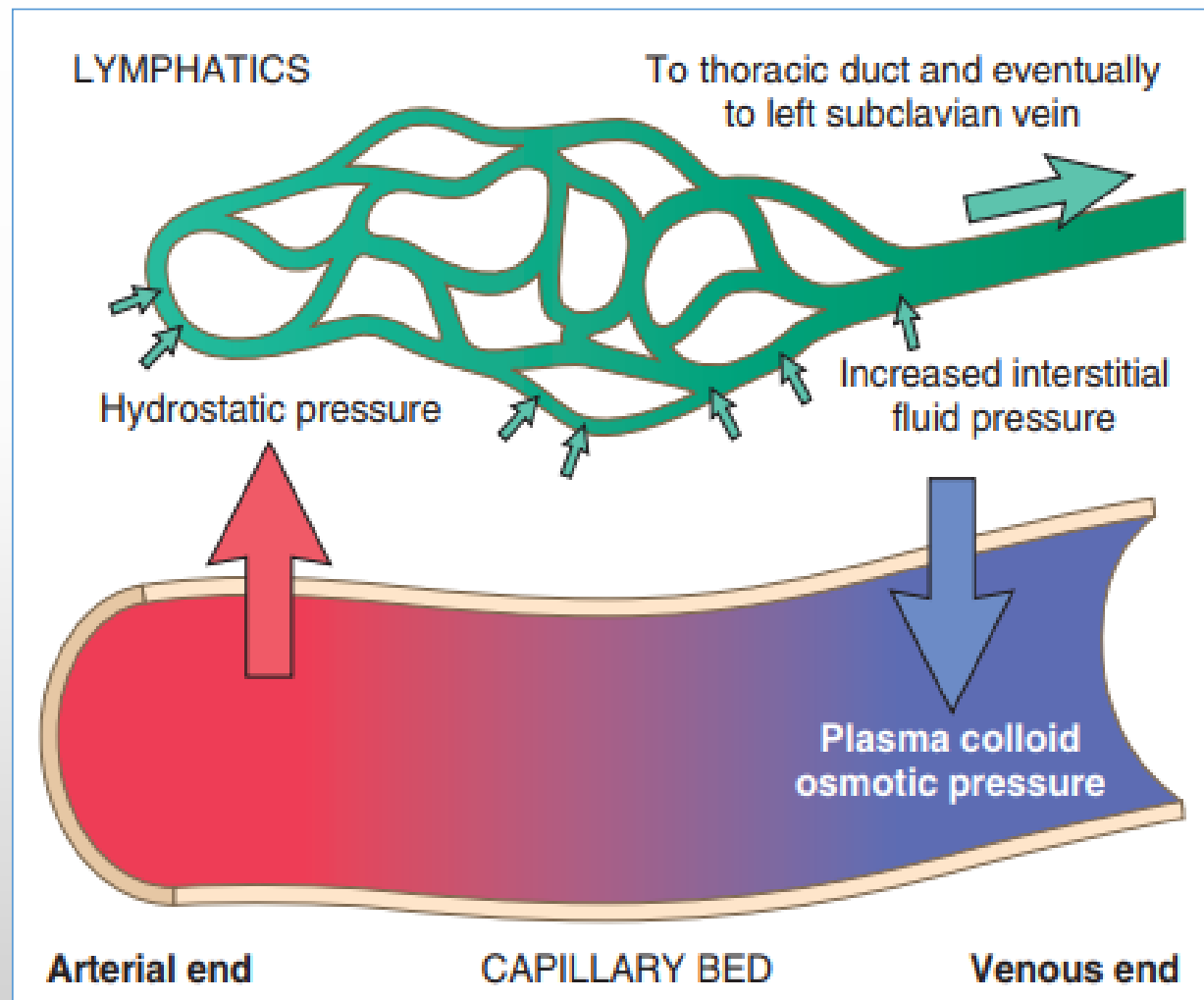
Pitting edema

Types of edema

1) **Inflammatory edema**, is a protein-rich **exudate** with high specific gravity, occurs due to increased vascular permeability.

2) **Non inflammatory edema**, is a protein-poor **transudate** that has low specific gravity, occurs due to other reasons other than inflammation.





Causes of Edema

- 1) Increased hydrostatic pressure (e.g., heart failure)
- 2) Increased vascular permeability (e.g., inflammation)
- 3) Decreased colloid osmotic pressure resulting from reduced plasma albumin
- 4) Decreased synthesis (e.g., liver disease, protein malnutrition)
- 5) Increased loss (e.g., nephrotic syndrome)
- 6) Lymphatic obstruction (e.g., inflammation or neoplasia)
- 7) Sodium retention (e.g., renal failure)

Hemorrhage

- Hemorrhage: the extravasation of blood from vessels, is most often the result of damage to blood vessels or defective clot formation.
- Capillary bleeding can occur in chronically congested tissues. Trauma, inflammatory or neoplastic erosion of a vessel wall also may lead to hemorrhage, which may be extensive if the affected vessel is a large vein or artery.



Types of Hemorrhage

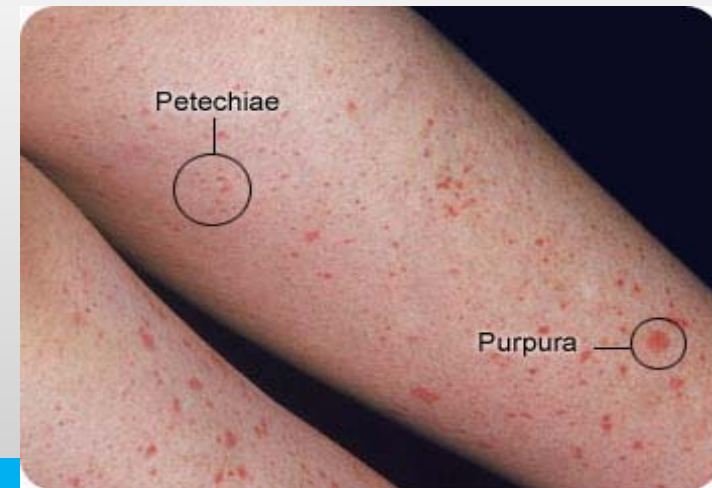
Hemorrhage may be external or accumulate within a tissue as a hematoma .

Hematoma: Accumulated hemorrhage which confined within a tissue, it could be insignificant (e.g., a bruise) or can involve so much bleeding as to cause death (e.g., rupture of aortic artery) .



2- **Petechiae**: Minute (1- 2-mm) hemorrhages into skin, mucous membranes, or serosal surfaces, typically associated with locally increased intravascular pressure, low platelet counts (thrombocytopenia), defective platelet function, or clotting factor deficiencies.

3- **Purpura**: Slightly larger (3- 5-mm) hemorrhages can be associated with many of the same disorders that cause petechiae; in addition, purpura can occur with trauma, vascular inflammation (vasculitis).



- **Ecchymoses**: Larger (1- 2-cm) subcutaneous hematomas (bruises). The erythrocytes in these local hemorrhages are phagocytosed and degraded by macrophages; the hemoglobin (red-blue color) is enzymatically converted into bilirubin (blue-green color) and eventually into hemosiderin (golden brown), accounting for the characteristic color changes in a hematoma.



- **Large bleeds into body cavities** are described variously according to location: hemothorax, hemopericardium, hemoperitoneum, or hemarthrosis (in joints).

